

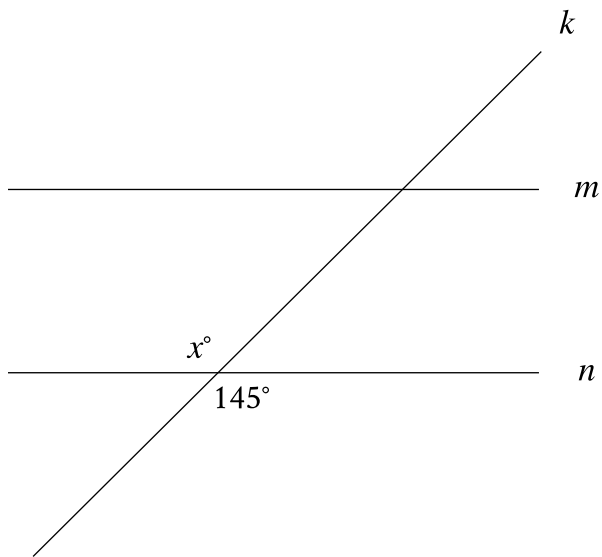
● EASY

● GEOMETRY AND TRIGONOMETRY

Lines, angles, and triangles

10 questions · ~15 min

02



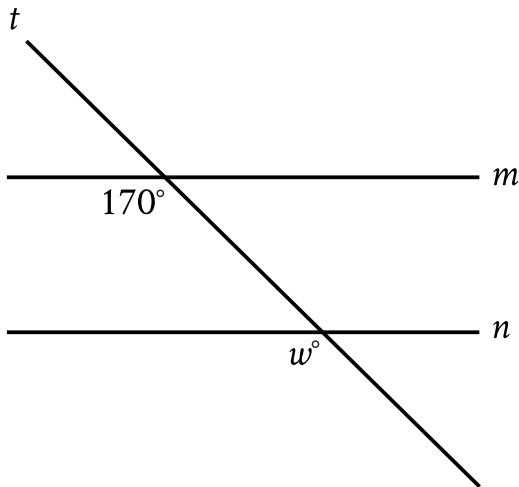
Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled k , m , and n .
- Line k intersects both line m and line n .
- At the intersection of line k and line n , 2 angles are labeled clockwise from top left as follows:
 - Top left: x°
 - Bottom right: 145°
- A note indicates the figure is not drawn to scale.

In the figure, line m is parallel to line n , and line k intersects both lines. Which of the following statements is true?

- A. The value of x is less than 145.
- B. The value of x is greater than 145.
- C. The value of x is equal to 145.

D. The value of x cannot be determined.

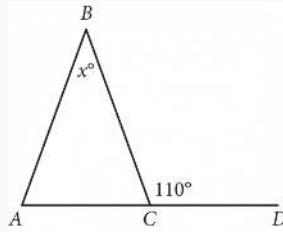


Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t , m , and n .
- Line t intersects both line m and line n .
- At the intersection of line t and line m , 1 angle is labeled clockwise from top left as follows:
 - Bottom left: 170°
- At the intersection of line t and line n , 1 angle is labeled clockwise from top left as follows:
 - Bottom left: w°
- A note indicates the figure is not drawn to scale.

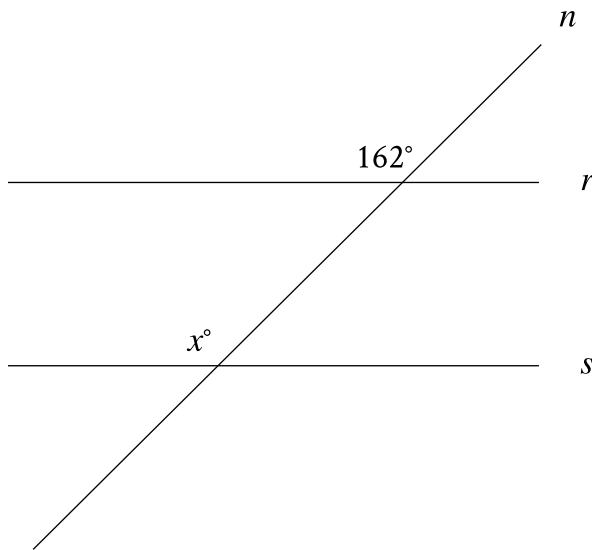
In the figure, line m is parallel to line n . What is the value of w ?

- A. 17
- B. 30
- C. 70
- D. 170



In the given figure, $\overline{AC}$ extends to point D. If the measure of $\angle BAC$ is equal to the measure of $\angle BCA$, what is the value of x ?

- A. 110
- B. 70
- C. 55
- D. 40



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled n , r , and s .
- Line n intersects both line r and line s .
- At the intersection of line n and line r , 1 angle is labeled clockwise from top left as follows:
 - Top left: 162°
- At the intersection of line n and line s , 1 angle is labeled clockwise from top left as follows:
 - Top left: x°
- A note indicates the figure is not drawn to scale.

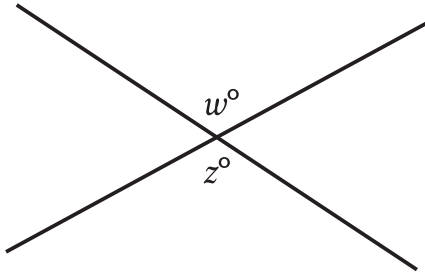
In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0

$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$
$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

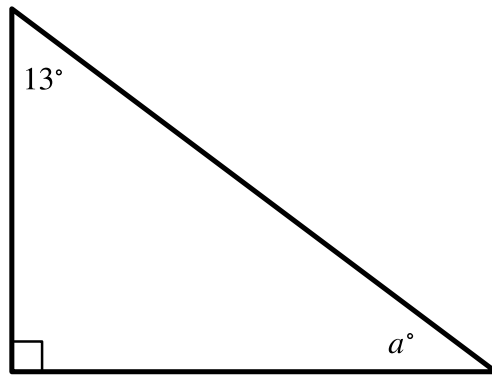


Note: Figure not drawn to scale.

- At the intersection of the 2 lines, the angles are labeled clockwise from top as follows:
 - Top: w°
 - Right: unlabeled
 - Bottom: z°
 - Left: unlabeled
- A note indicates the figure is not drawn to scale.

In the figure, two lines intersect at a point. If $w = 136$, what is the value of z ?

- A. 36
- B. 44
- C. 68
- D. 136

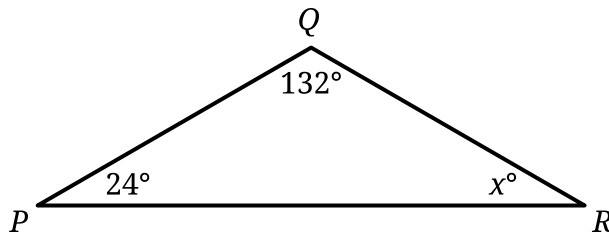


Note: Figure not drawn to scale.

- One angle is a right angle.
- The measures of the other 2 angles are as follows:
 - a°
 - 13°
- A note indicates the figure is not drawn to scale.

In the right triangle shown, what is the value of a ?

- A. 13
- B. 77
- C. 90
- D. 103



Note: Figure not drawn to scale.

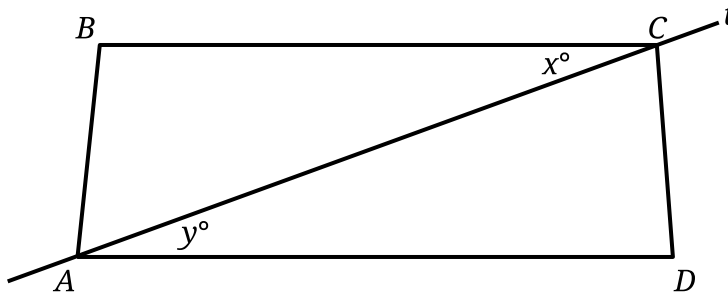
- The measure of angle upper P is 24° .
- The measure of angle upper Q is 132° .
- The measure of angle upper R is x° .
- A note indicates the figure is not drawn to scale.

In the triangle shown, $PQ = QR$. What is the value of x ?

- A. 156
- B. 66
- C. 48
- D. 24

In triangle ABC , $AB = 4,680$ millimeters mm and $BC = 4,680$ mm. Which statement is sufficient to prove that triangle ABC is equilateral?

- A. $AC = 4,680$ mm
- B. $AC = 468$ mm
- C. $AC = 46.8$ mm
- D. $AC = 4.68$ mm



Note: Figure not drawn to scale.

- Line t passes through points upper A and upper C.
- The measure of angle upper A upper C upper B is x° .
- The measure of angle upper C upper A upper D is y° .
- A note indicates the figure is not drawn to scale.

In the figure shown, $\overline{BC}$ is parallel to $\overline{AD}$ and $x = 22$. Line t passes through points A and C . What is the value of y ?

- A. 22
- B. 44
- C. 68
- D. 158

Triangles EFG and JKL are congruent, where E , F , and G correspond to J , K , and L , respectively. The measure of angle E is 45° and the measure of angle F is 20° . What is the measure of angle J ?

- A. 20°
- B. 45°
- C. 135°
- D. 160°